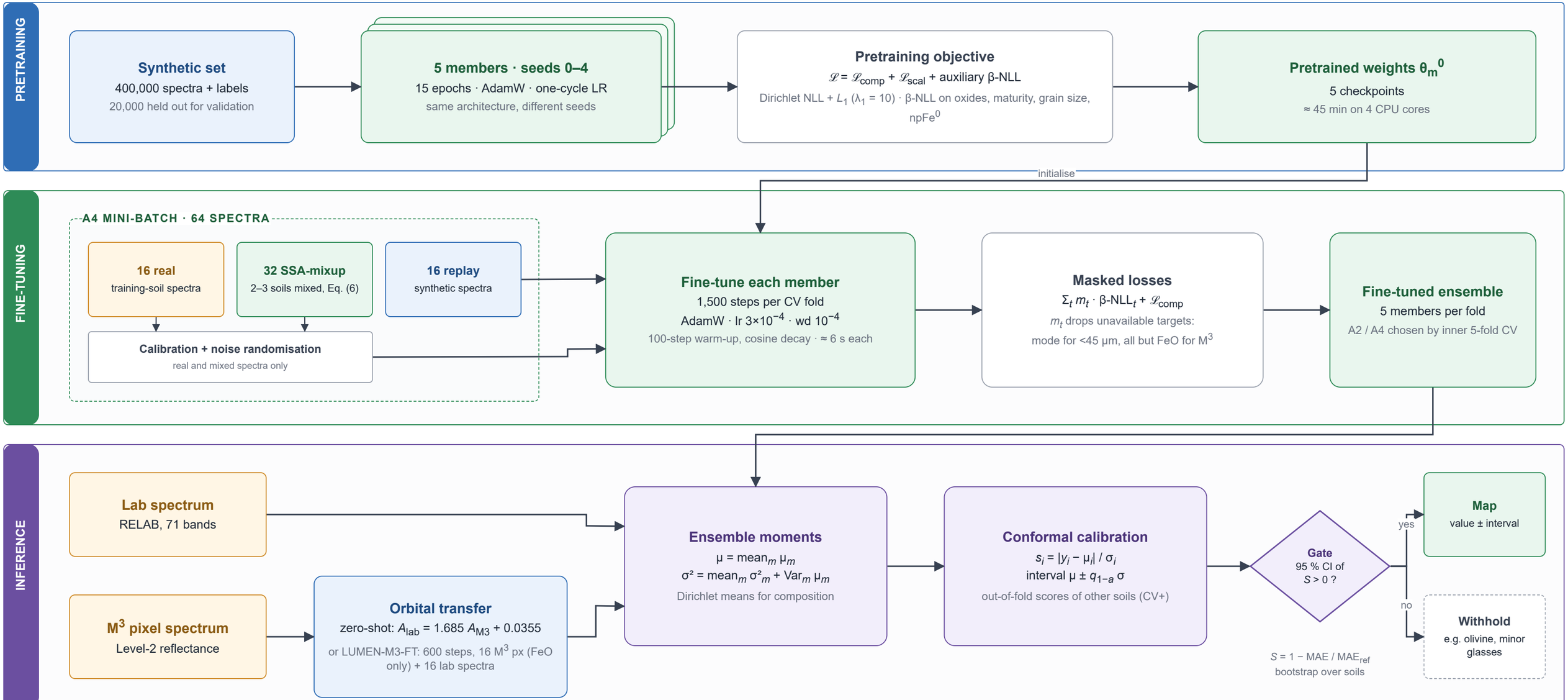

PRETRAINING
Synthetic set
400,000 spectra + labels
20,000 held out for validation
5 members · seeds 0–4
15 epochs · AdamW · one-cycle LR
same architecture, different seeds
Pretraining objective
$\mathscr{L} = \mathscr{L}_{comp} + \mathscr{L}_{scal}$ + auxiliary β-NLL
Dirichlet NLL + L_1 ($\lambda_1 = 10$) · β-NLL on oxides, maturity, grain size, npFe^0
Pretrained weights θ_m^0
5 checkpoints
≈ 45 min on 4 CPU cores
initialise
FINE-TUNING
A4 MINI-BATCH · 64 SPECTRA
16 real
training-soil spectra
32 SSA-mixup
2–3 soils mixed, Eq. (6)
16 replay
synthetic spectra
Calibration + noise randomisation
real and mixed spectra only
Fine-tune each member
1,500 steps per CV fold
AdamW · lr 3×10^{-4} · wd 10^{-4}
100-step warm-up, cosine decay · ≈ 6 s each
Masked losses
$\Sigma_t\, m_t \cdot \beta\text{-NLL}_t + \mathscr{L}_{comp}$
m_t drops unavailable targets:
mode for <45 µm, all but FeO for M^3
Fine-tuned ensemble
5 members per fold
A2 / A4 chosen by inner 5-fold CV
INFERENCE
Lab spectrum
RELAB, 71 bands
M^3 pixel spectrum
Level-2 reflectance
Orbital transfer
zero-shot: $A_{lab} = 1.685\, A_{M3} + 0.0355$
or LUMEN-M3-FT: 600 steps, 16 M^3 px (FeO only) + 16 lab spectra
Ensemble moments
$\mu = \text{mean}_m\, \mu_m$
$\sigma^2 = \text{mean}_m\, \sigma^2_m + \text{Var}_m\, \mu_m$
Dirichlet means for composition
Conformal calibration
$s_i = |y_i - \mu_i| / \sigma_i$
interval $\mu \pm q_{1-a}\, \sigma$
out-of-fold scores of other soils (CV+)
Gate
95 % CI of $S > 0$?
yes
no
$S = 1 - MAE / MAE_{ref}$
bootstrap over soils
Map
value ± interval
Withhold
e.g. olivine, minor glasses